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Review

Impacts of nitrogen pollution on corals in the context of global climate change and potential strategies to conserve coral reefs

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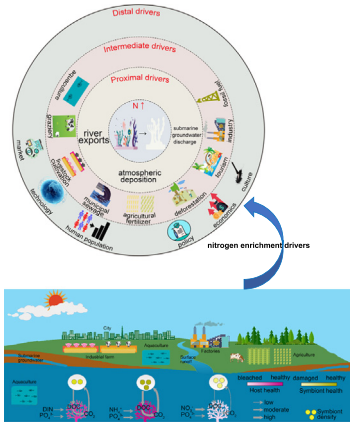
HIGHLIGHTS

- Anthropogenic nitrogen and phosphorus increase in the oceans and seas.
- Nitrogen can directly affect corals, or indirectly through algae and viruses.
- Nitrate enrichment and ammonium enrichment have different effects on corals.
- The tolerance thresholds for nitrogen can greatly differ among corals.
- Reducing nitrate input and CO_2 emissions are promising conservation strategies.

GRAPHICAL ABSTRACT

NOTE: Correlation between coral and symbiodiniaceans is partially referenced to the literature (Morris et al. 2019).



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ABSTRACT

Ocean warming has severe impacts on coral reef ecosystems with frequent incidences of coral bleaching. In addition, eutrophication poses an increasing threat to coral reef environments and has been found to increase the vulnerability of corals to thermal bleaching. Eutrophication has accelerated in recent years with coastal nutrient loads expected to continue to increase under global change. However, the mechanisms by which nutrient pollution affects corals and coral reefs are still under debate, in particular with regard to nitrogen. The main objective of this paper is to review mechanisms by which nitrogen pollution affects coral health and corresponding strategies to reduce the impact of nitrogen pollution. Different coral species possess varying tolerance thresholds for nitrogen enrichment and corals show differential responses to enrichment with nitrate and ammonium. Nitrate assimilation increases oxidative stress in corals, promotes growth of the phototrophic symbionts in corals, and induces phosphate starvation in these symbionts, which further impairs the symbiosis. In contrast, a moderate supply of ammonium is mostly beneficial for coral development. In addition, combined nitrogen and

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Coral bleaching

phosphorous enrichment can indirectly compromise coral health by enhancing macroalgae growth and increasing the incidence of coral diseases caused by predation on corals. It must be realized that both levels of nutrient pollution and the stoichiometric ratios of C: N: P: Fe availabilities determine the ultimate effect of nutrients on coral health. We confirm the strategy to conserve coral reefs via coral-targeted water quality management, in particular by including a reduction of the nitrate influx and by proper management of fish stocks to facilitate healthy reef ecology.

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1. Introduction

Coral reefs are indispensable ecosystems, providing resources and ecological services worth approximately $375 billion–$9.9 trillion USD per year for fisheries, tourism, raw materials, coastal protection and many more (Costanza et al. 1997, 2014). However, due to climate change and local impacts by human activities, one third of coral species are threatened with extinction (Carpenter et al. 2008). If business continues as usual, more than 90% of coral reefs are expected to be in danger of long-term annual degradation (Grottoli et al. 2014). Coral coverage has declined in the Great Barrier Reef by 52% in the period 1985–2012 (De'ath et al. 2012), by 63.4% in the Meso-American Reef between 1966 and 2015 (Bland et al. 2017), by 52% in the Florida Keys over the past 240 years (McClenachan et al. 2017), by 50% in the northern Caribbean Sea from 1985 to 2016 (Estrada-Saldivar et al. 2019), and by at least 80% on coastal fringing reefs along the Chinese mainland and adjoining Hainan Island since the 1980s (Hughes et al. 2013; Cybulski et al. 2020).

Human activities since the Industrial Revolution contributed to the rapid increase of greenhouse gas emissions. This has led to a rise in air and ocean temperatures that has not been experienced for at least the past fifty to hundreds of millions of years (Hönisch et al. 2012; Wright and Schaller 2013). Xu et al. (2018) predicted that the 1.5 °C increase in sea surface temperature (SST), projected for the year 2040 in the IPCC special report, could already be breached by 2030. Increasing SST leads to an increased frequency of coral bleaching, which is the mass expulsion of phototrophic symbionts (dinoflagellate algae belonging to the family Symbiodiniacea) by the coral host. Thermal bleaching is the response to heat stress (Hughes et al. 2017). Since the large-scale global coral bleaching event of 1983, the first mass bleaching event that was associated with rising SST (Gill et al. 2006; Spalding and Brown 2015), the frequency and severity of large-scale bleaching events have significantly increased (Hughes et al. 2017).

Increased input of inorganic nutrients, in particular dissolved inorganic nitrogen (DIN), has also been associated with reef decline. Numerous field observations and laboratory simulations have shown adverse effects of nitrogen enrichment on coral growth and calcification (Silbiger et al. 2018), coral heat resistance (Wooldridge et al. 2013), the resilience of individual species of corals (Hall et al. 2018), and incidence and severity of coral disease (Vega Thurber et al. 2014). Moreover, an interaction is suggested between enrichment with nutrients and other detrimental stressors such as elevated temperature, ocean acidification and overfishing (DeCarlo et al. 2015; Zaneveld et al. 2016; Donovan et al. 2020). In an effort to control those adverse impacts, governments want to develop policies to manage water quality and maintain the valuable functions of coral reef ecosystems. To be able to specify water quality standards, however, the lowest adverse effect levels of nitrogen and other nutrients need to be known. This paper reviews the current status of nitrogen pollution in coral environments, and summarizes the different types of impact of different forms of nitrogen pollution on corals and the modulation of these effects by other nutrients such as phosphorous. Finally, we propose measures to improve the nitrogen-related water quality status by focusing on the sources of nitrogen pollution and mitigation options based on known impact mechanisms.

2. Marine nitrogen pollution

Since the Industrial Revolution, global anthropogenic inputs of reactive nitrogen into the biosphere have increased from ~15 Tg year^{-1} in 1850 to ~235 Tg year^{-1} in 2020 (Penuelas et al. 2020). Among the approximately 13.5-fold increase in anthropogenic nitrogen, nitrogen fertilizer contributed the most, from ~1 Tg year^{-1} in 1850 to ~116 Tg year^{-1} in 2020 (Penuelas et al. 2020). The increased use of nitrogen fertilizer ensures that grain production is synchronized with global population growth (Steffen et al. 2011). However, most cereal crops require no more than 50% of the supplied nitrogen fertilizer (Hakeem et al. 2017), resulting in large amounts of excess nitrogen, which are transported to the ocean through rivers (Strokal et al. 2016b) or volatilize into the atmosphere. Other important drivers causing anthropogenic nitrogen increase are municipal sewage and industrial farm sewage (Wear and Thurber 2015; Strokal et al. 2017; Barile 2018), factories (including chemical industry and fossil-fuel emissions) (Penuelas et al. 2020), and aquaculture (Herbeck et al. 2014; Wang et al. 2020a) (Fig. 1).

Concurrent with the increase of anthropogenic nitrogen, the nitrogen levels in the ocean also increased. The amount of dissolved inorganic nitrogen (DIN) transported to ocean through rivers was ~19 Tg year^{-1} in the year 2000 and is predicted to be ~25 Tg year^{-1} in

the year 2050 (Seitzinger et al. 2010). Deposition of atmospheric nitrogen was ~41 Tg year^{-1} in 2005 and is predicted to be ~39 Tg year^{-1} in 2050 (Okin et al. 2011; Kanakidou et al. 2016). Input of nitrogen through fresh submarine groundwater discharge (SGD) was ~1.4 Tg year^{-1} in 2000 and was predicted to increase to ~1.7 Tg year^{-1} in 2050 (Beusen et al. 2013). Total SGD-derived DIN input into the oceans, including the discharge of saline and fresh groundwater as well as pore water advection, was estimated to be ~32 Tg year^{-1} in 2000 (Beusen et al. 2013). It far exceeds the contribution of fresh SGD to the input of nitrogen into marine systems and even the contribution of river exports. It has been postulated that a significant fraction of DIN originates from the remineralization of organic matter stored within the sediments (Cho et al. 2018; Weinstein et al. 2011). However, this increase in oceanic nitrogen is not accompanied by a proportional increase in phosphorus (Mahowald et al. 2008; Seitzinger et al. 2010; Vilmin et al. 2018), which causes an increase in the ratios between nitrogen and phosphorus in the ocean (Fig. 1).

Due to the above-mentioned reasons, many coral reef areas are subjected to nutrient enrichment, especially with nitrogen. In Asian countries, chronic nutrient enrichment is often caused by marine cage aquaculture (Huang et al. 2011; Herbeck et al. 2014; Wang et al. 2020a). It was also estimated that in Chinese rivers, 20–78% of the dissolved nitrogen and phosphorous originates from animal manure. This is caused by the disconnect between China's industrialized livestock and crop production systems (Strokal et al. 2016a). With population increase and urbanization, the effects of this disconnection on rivers and hence, on coastal waters, may further increase in the future.

Volatile ammonium and nitrogen oxides from nitrogen fertilizers and fossil fuels increase the amount of reactive nitrogen in the atmosphere (Holtgrieve et al. 2011), thereby increasing atmospheric nitrogen deposition. Atmospheric deposition not only affects the nitrogen cycle and ecosystems in coastal waters (Zhu et al. 2013), but also affects remote sea ecosystems (Ren et al. 2017), which can be attributed to the efficient transport of nitrogen through atmospheric circulation (Duce et al. 2008). From a global perspective, the contribution of atmospheric deposition to the nitrogen content in the ocean has even exceeded that of river transportation (Fig. 1). Although in some areas, such as the North Atlantic (Wang et al. 2018), the impact of atmospheric nitrogen deposition on the marine ecological environment is relatively small due to severe land-based sources of nutrient pollution (Menesguen and Lacroix 2018).

Submarine groundwater discharge (SGD) is a less obvious, but still relevant pathway to increase marine nutrient levels (Adyasari et al. 2019; Oehler et al. 2019). The main pathways of nitrogen entering coastal areas by SGD are through tidal systems and small deltas (Beusen et al. 2013; Babu et al., 2021). The nitrate transport through these types of SGD is spread all over the world and can be found in Southeast Asia (Oehler et al. 2018), South Pacific (Erler et al. 2018), and the United States (Amato et al. 2016; Fackrell et al. 2016; Prouty et al. 2017).

3. Effects of nitrogen enrichment on corals

Nitrogen enrichment can directly affect coral physiology and can indirectly affect coral reef resilience through promoting the growth of other organisms. Direct effects on coral physiology include a reduced heat tolerance (Wooldridge et al. 2013), longer bleaching duration (Burkepile et al. 2019) and reduced rates of calcification (Kinsey and Davies 1979; Silbiger et al. 2018). Tolerance thresholds to nitrogen enrichment are genus- to species- and symbiont-specific (Table 1 and Fig. 2) (Duprey et al. 2016; Ezzat et al. 2017; Cybulski et al. 2020). Indirect effects on coral reef resilience include the induction of diseases through the stimulation of viruses (Vega Thurber et al. 2017) and pathogenic bacteria (Lesser et al. 2007; Barott and Rohwer 2012), and an enhanced growth of competitors for light and space such as macroalgae (De'ath and Fabricius 2010) and phytoplankton (D'Angelo and Wiedenmann 2014). The proliferation of phytoplankton due to nitrogen enrichment can also increase the cloning frequency and developmental rate of the crown-of-thorns starfish larvae (*Acanthaster planci*, natural predators of corals), thereby increasing coral predation (Allen et al. 2019).

It should be noted here that nitrogen is not a stand-alone nutrient, but that it acts in conjunction with phosphate (P), ferrum (Fe), and other micronutrients (Roff and Mumby 2012; Wang et al. 2019). Effects of multiple-nutrient enrichment on the prevalence of coral competitors and coral diseases will be reviewed in Section 3.1. In addition, evidence has accumulated in recent years that the responses of coral physiology to enrichment with ammonium are fundamentally different from responses to enrichment with nitrate (Shantz and Burkepile 2014; Morris et al. 2019). To better understand this disparity, Sections 3.2 and 3.3 summarize the differential effects of nitrate and ammonium on two key factors related to coral reef resilience: the sensitivity to thermal bleaching and the rate of calcification.

3.1. Nutrient enrichment and coral diseases

Nutrient enrichment can increase the incidence and severity of coral diseases (Fig. 3). Several types of diseases have been found to increase in prevalence under excessive nutrient conditions, such as white spot disease (Kaczmarsky 2006), white pox (Patterson et al. 2002), black band disease (Voss and Richardson 2006), aspergillosis and yellow blotch diseases (Bruno et al. 2003).

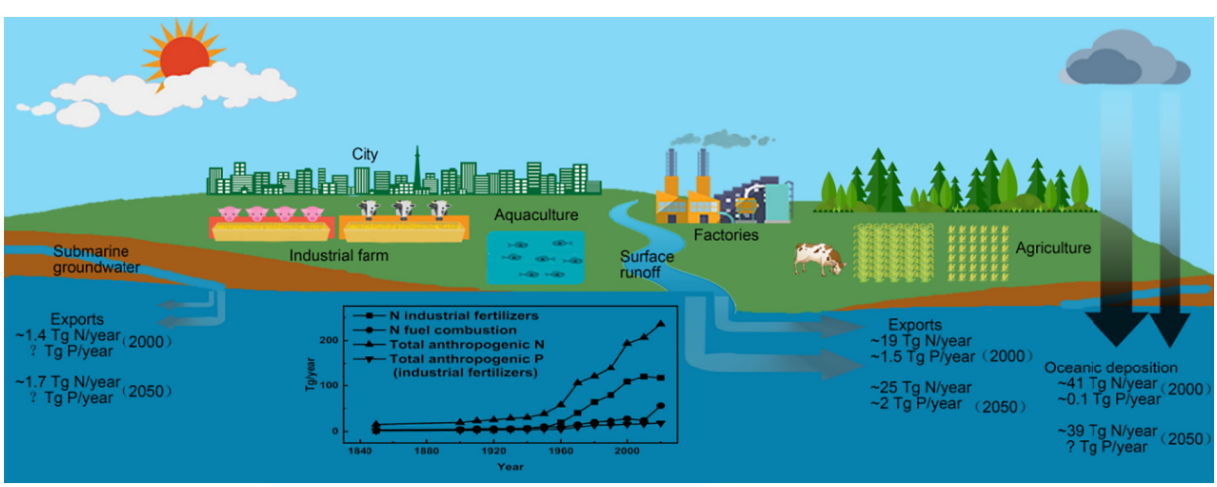

Fig. 1. Past and future status of worldwide marine nutrient pollution.

Table 1
Individual coral species' response to elevated nutrient concentrations as determined in a laboratory experiment (L), an in-situ experiment (IE) or via in-situ data collection (ID). The factors that the study focused on were nitrogen (N); phosphate (P); temperature (T); ocean acidification (OA) and/or light intensity (LI); DSS = Dark Spot Syndrome, Fv/Fm = A quantity related to the maximum quantum yield of PSII photochemistry.

Survey site	Species	Data sources	Factors	Concentration	Response	References
Red Sea	*Stylophora pistillata*	L (9 weeks)	N	PO_4^{3-} = 0.05 μM, NO_3^- = 0.5 μM, NH_4^+ = 2.5 μM	Carbon fixation and symbiont to host translocation increased	Ezzat et al. (2015)
Red Sea	*Stylophora pistillata*	L (9 weeks)	N	PO_4^{3-} = 0.05 μM, NO_3^- = 3 μM	Carbon fixation and symbiont to host translocation decreased	Ezzat et al. (2015)
Red Sea	*Stylophora pistillata*	L (9 weeks)	N,P	PO_4^{3-} = 1 μM, NO_3^- = 0.5 μM, NH_4^+ = 2.5 μM	Symbiodiniceans densities decreased; nutrient allocation shifted to symbionts	Ezzat et al. (2015)
Red Sea	*Stylophora pistillata*	L (6 weeks)	N	NH_4^+ = 4 μM	Chlorophyll a and total protein increased, photosynthesis remained stable	Bednarz et al. (2020)
Mediterranean	*Cladocora caespitosa*	L (6 weeks)	N	NO_3^- = 6–7 μM, NH_4^+ = 5–6 μM, T = 17 °C (normal growth temperature)	Symbiont density and gross photosynthesis decreased, calcification, respiration and net photosynthesis maintained	Hadjioannou et al. (2019)
Mediterranean	*Cladocora caespitosa*	L (6 weeks)	N,T	NO_3^- = 6–7 μM, NH_4^+ = 5–6 μM, T = 24 °C	Symbiont density and respiration maintained, calcification, net photosynthesis and gross photosynthesis increased	Hadjioannou et al. (2019)
Hawaii	*Porites lobata*	ID (2013.07)	N,OA	NO_3^- = 0.16 ± 0.10 μM pH = 8.00 ± 0.02	Growth rate = 0.88 ± 0.06 $cm \cdot yr^{-1}$ calcification rates = 0.94 $g \cdot cm^{-2} \cdot yr^{-1}$ bioerosion rate = 56.03 $mg \cdot cm^{-2} \cdot yr^{-1}$	Prouty et al. (2017)
Hawaii	*Porites lobata*	ID (2013.07)	N,OA	NO_3^- = 0.41 ± 0.18 μM pH = 8.01 ± 0.03	Growth rate = 0.72 ± 0.16 $cm \cdot yr^{-1}$ calcification rates = 0.67 $g \cdot cm^{-2} \cdot yr^{-1}$ bioerosion rate = 39.87 $mg \cdot cm^{-2} \cdot yr^{-1}$	Prouty et al. (2017)
Hawaii	*Porites lobata*	ID (2013.07)	N,OA	NO_3^- = 0.19 ± 0.11 μM pH = 8.01 ± 0.02	Growth rate = 0.95 ± 0.11 $cm \cdot yr^{-1}$ calcification rates = 1.02 $g \cdot cm^{-2} \cdot yr^{-1}$ bioerosion rate = 22.58 $mg \cdot cm^{-2} \cdot yr^{-1}$	Prouty et al. (2017)
Hawaii	*Porites lobata*	ID (2013.07)	N,OA	NO_3^- = 20.35 ± 23.32 μM pH = 7.85 ± 0.17	Growth rate = 0.69 ± 0.10 $cm \cdot yr^{-1}$ calcification rates = 0.68 $g \cdot cm^{-2} \cdot yr^{-1}$ bioerosion rate = 99.15 $mg \cdot cm^{-2} \cdot yr^{-1}$	Prouty et al. (2017)
Florida Keys	*Siderastrea siderea*	IE (2009.06-2012.06)	N,P	DIN = 3.91 ± 1.34μM, DIP = 0.27 ± 0.07 μM	The prevalence of DSS in corals doubled (22%) and severity of DSS significantly increased	Vega Thurber et al. (2014)
Florida Keys	*Siderastrea radians*	IE (2009.06-2012.06)	N,P	DIN = 3.91 ± 1.34 μM, DIP = 0.27 ± 0.07 μM	DSS prevalence = 23%	Vega Thurber et al. (2014)
Florida Keys	*Stephanocoenia intersepta*	IE (2009.06-2012.06)	N,P	DIN = 3.91 ± 1.34 μM, DIP = 0.27 ± 0.07 μM	DSS prevalence = 7%	Vega Thurber et al. (2014)
Florida Keys	*Agaricia* spp.	IE (2009.06-2012.06)	N,P	DIN = 3.91 ± 1.34 μM, DIP = 0.27 ± 0.07 μM	DSS prevalence = 1%, The prevalence of coral bleaching increased 3–6 times	Vega Thurber et al. (2014)
Great Barrier Reef	*Pocillopora damicornis*	IE (1995.01-1996.02)	N,P	PO_4^{3-} = 2.40 μM, NH_4^+ = 11.30 μM	Coral mortality increased about 2 times	Koop et al. (2001)
Great Barrier Reef	*Acropora longicyathus*	IE (1995.01-1996.02)	N,P	NH_4^+ = 11.30 μM	No effect on calcification rate, increased coral mortality.	(Koop et al., 2001)
Great Barrier Reef	*Acropora longicyathus*	IE (1995.01-1996.02)	N,P	PO_4^{3-} = 2.40 μM	Coral skeleton linear extension rate increased by 1/4	Koop et al. (2001)
Great Barrier Reef	*Acropora longicyathus*	IE (1995.01-1996.02)	N,P	PO_4^{3-} = 2.40 μM, NH_4^+ = 11.30 μM	Increased coral skeleton micro-density significantly, increased coral mortality.	Koop et al. (2001)
Southampton	*Euphyllia paradivisa*	L (9 weeks)	N,LI	DIN = 3 μM, DIP = 0.07 μM, light levels = 160 μmol $m^{-2}s^{-2}$	Fv/Fm dropped, strongly bleaching, chlorophyll a reduced	Wiedenmann et al. (2013)

Three pathways for this increased incidence of coral disease under nutrient enrichment have been proposed (Fig. 3). The first pathway is a direct increase in the abundance of coral-associated eukaryote (predominantly herpes-like) viruses under high nutrient loadings (Vega Thurber et al. 2008). The second pathway is termed the "virus-mediated vortex of coral reef decline" (Vega Thurber et al. 2017). This Viral Vortex represents a nutrient-induced positive feedback loop where eutrophication, in conjunction with heat stress, promotes the occurrence of both eukaryote viruses and bacteriophages in corals. Increased infection of the entire coral holobiont with viruses causes the corals to increase their loss of organic carbon and nitrogen via excretion of mucus and subsequently stimulates microbial growth in the water column. Abundance of water column viruses then rises as well, causing lysis of water-column microbes and a subsequent transfer of nutrients back to the benthic system. This virus-mediated flow of nutrients towards the benthos then further promotes coral disease and hence, reef decline (Vega Thurber et al. 2017).

The third pathway of nutrient-induced coral disease, which is likely to be interlinked with the Viral Vortex, entails a sequence of events known as DDAM (Dissolved organic matter, Disease, Algae, Microbes) (Barott and Rohwer 2012). The DDAM pathway links increased incidence of coral disease to an increased dominance of algae (including turf algae and fleshy algae) and cyanobacteria on nutrient-enriched reefs. Algae and cyanobacteria adapt to higher nutrient environments faster than corals and multiply more rapidly under eutrophic conditions (McCook 1999; Zaneveld et al. 2016). Consequently, the release of dissolved organic matter by algae increases, which increases the C and N content in the benthic environment, which may further lead to the transfer of coral reefs to benthic cyanobacteria mats or algae-based systems, thereby forming a negative feedback loop (Fig. 3) (Brocke et al. 2018; Ford et al. 2018). In addition, it also can induce changes in the coral microbiome towards higher proportions of putative pathogens and hence, a higher incidence and severity of coral diseases (Kuntz et al. 2005; Kline et al. 2006; Smith et al. 2006). Similar to the Viral Vortex, DDAM is a positive feedback loop: the higher the coral mortality rate, the more space becomes available for algae to continue the DDAM dynamics (Dinsdale and Rohwer 2011). Although the ecological relevance of the DDAM pathway is not fully understood, the pathway has repeatedly been recognized (Kuntz et al. 2005; Kline et al. 2006; Smith et al. 2006; Dinsdale and Rohwer 2011; Haas et al. 2013; Zaneveld et al. 2016) and deserves further investigation.

It should be noted that effects of nutrients on coral disease may differ per location. For example, the algae-mediated DDAM pathway may not occur in areas where addition of nutrients does not stimulate algal

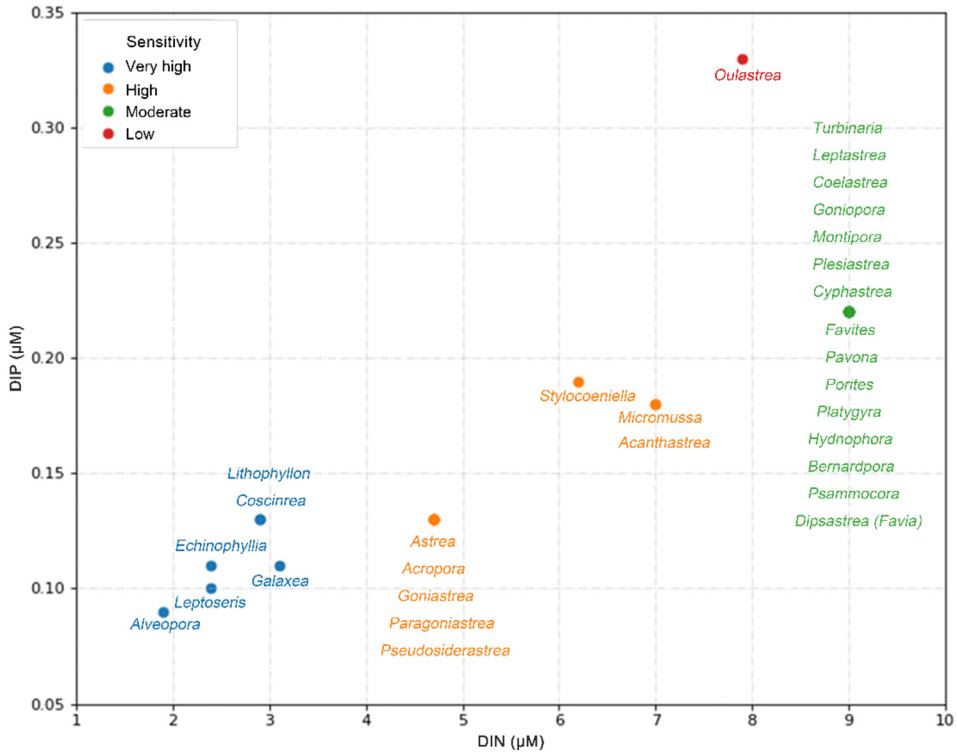

Fig. 2. Thresholds of nitrogen and phosphorus tolerance in corals of different genera.

growth because of strong herbivorous controls on macroalgae (e.g. Rasher et al., 2012) or due to limited availability of particular micronutrients. It has been suggested that the availability of iron (Fe) causes disparities in nutrient-effects among different geographical regions. Where iron is limiting macroalgae growth, additions of other nutrients will not notably affect coral-macroalgae competition (Roff and Mumby 2012). Hence, studies on the effects of nutrients on coral-macroalgae competition and associated aspects of reef resilience should

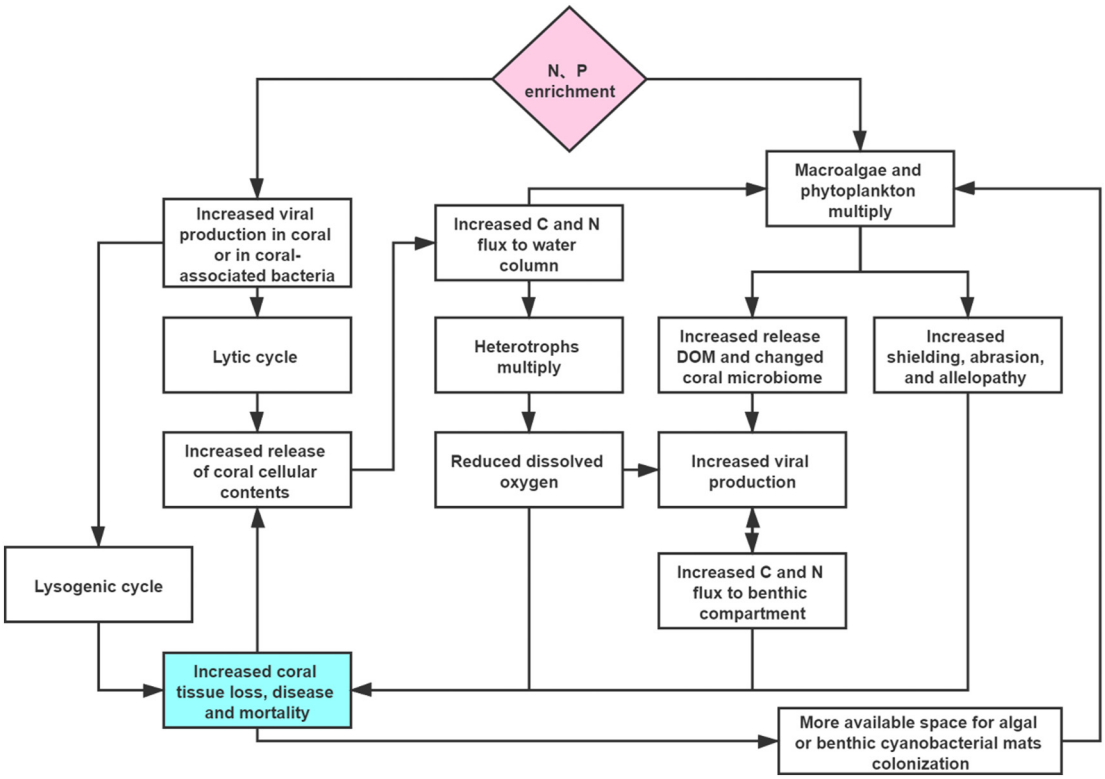

Fig. 3. Relationship between nutrient enrichment and coral diseases.

aim to distinguish between individual nutrients (e.g. Fe, P, N) and the forms in which these nutrients appear (e.g. nitrate versus ammonium).

In general, there is little information on the roles that the individual nutrient components (e.g. nitrate, ammonium, phosphate) play in relation to causation of coral disease. As mentioned previously, nitrate and ammonium show differential effects on other aspects of coral biology: addition of nitrate is detrimental to calcification and photosynthesis, whereas moderate additions of ammonium benefit both processes (Shantz and Burkepile 2014; Fernandes de Barros Marangoni et al., 2020). However, most studies that experimentally investigated the relationship between nutrients and coral disease (e.g. Bruno et al. 2003; Vega Thurber et al. 2014; Zaneveld et al. 2016) used commercial fertilizers that were composed of a balanced mix of nitrogen (supplied as nitrate and ammonium), phosphorous and micronutrients. As such, these studies could not discriminate between effects of individual nutrient components. Nevertheless, some preliminary conclusions can be made. Increased incidence of coral disease has been associated in particular with human-related sources of eutrophication such as sewage discharge or runoff of agricultural fertilizers. For example, stable isotope analysis indicated that at the island of Guam, the input of nitrogen from the local sewage discharge was significantly related to the severity of coral disease (Redding et al. 2013). Human-related sources of eutrophication are usually relatively high in nitrate (Shantz and Burkepile 2014), which suggests that the presence of nitrate outweighs potential benefits of ammonium with regard to causation of diseases. The results obtained from enrichment experiments with nitrate-rich fertilizers (Bruno et al. 2003; Vega Thurber et al. 2014; Zaneveld et al. 2016) support this view. For example, artificial addition of fertilizer increased the disease prevalence in *Siderastrea siderea* by 170% in the Florida Keys (Vega Thurber et al. 2014). *S. siderea* in Florida is currently projected to be declining at a rate of 1% per year due to black band disease, but chronic nutrient exposure from experiments caused an annual loss rate of 3%. Ten months after the termination of the nutrient enrichment experiment, the prevalence of coral disease in the treatment group was similar to that of the control group, indicating that reducing inorganic nutrient pollution may be an important way to lower the coral disease incidence. The exact mechanism underlying this example of nutrient-induced coral disease remains uncertain: nutrient enrichment may induce or exacerbate the symptoms of coral physiological diseases by reducing the threshold of coral resistance to pathogens, by increasing the pathogenicity of pathogens or both (Vega Thurber et al. 2014). In addition, nutrient-induced disturbance of the coral microbiome can turn allies into enemies: under nutrient enrichment, parrotfish bites (normally regarded as positive for reef development) cause an increased susceptibility of the bitten corals to pathogenic bacteria (Zaneveld et al. 2016).

3.2. Nitrogen enrichment and the sensitivity of corals to thermal bleaching

Coral bleaching is a term used to describe the loss of coloration in corals associated with the release of the symbiotic algae from the coral host (Fitt et al. 2001). The primary cause of bleaching is generally assumed to be oxidative stress due to the production of reactive oxygen species (ROS) under conditions where photosynthesis is compromised. ROS-induced bleaching happens most frequently under conditions of heat stress in combination with high illumination (Weis 2008) and is usually preceded by an imbalance in host-symbiont resource dynamics that drives the mutualistic character of the symbiosis towards a more commensalistic character at the expense of the coral host (Wooldridge 2013; Morris et al. 2019). The incidence and severity of this so-called thermal bleaching has been linked to increased inputs of inorganic nutrients on coral reefs (Wooldridge and Done 2009; Wiedenmann et al. 2013; Wooldridge et al. 2013). There are two interlinked mechanisms by which nutrient additions may reduce the threshold temperature for thermal bleaching: (1) nutrients may cause excessive growth of the symbionts above the threshold value for a stable symbiosis, and (2) unfavorable nutrient ratios due to selective enrichment with nitrogen, in particular nitrate, can compromise symbiodiniacean photosynthesis. Both mechanisms will be further outlined below.

The classical view on the effects of nutrient enrichment on corals is that nutrients stimulate the growth of symbionts *in hospite*, resulting in higher symbiont densities (Muscatine et al. 1989; Stambler et al. 1991). Under oligotrophic conditions, the growth of symbiotic algae *in hospite* is assumed to be limited by nutrients, in particular nitrogen (Krueger et al. 2020). When additional nutrients become available, the symbionts start to invest more of the photosynthetically derived metabolic energy in their own growth, which leads to higher symbiont densities and a reduced translocation of organic carbon to the host (Marubini and Davies 1996).

There is an optimal symbiont density at which the coral host benefits most from translocation of excess photosynthetic carbon products, reflected by the highest ratio between symbiont photosynthesis and holobiont respiration (Hoogenboom et al. 2010; Wooldridge 2017). Notwithstanding this, the symbiont density range that is suited for a mutualistic symbiosis is wide and varies among coral and symbiont species. This variability may explain why conflicting results have been reported with regard to nutrient enrichment, symbiont densities and sensitivity to bleaching (see review by Morris et al. 2019). A moderate increase in symbiont density (driven by moderate enrichment with nutrients) may further optimize the symbiosis and may be beneficial with respect to bleaching resistance (Ezzat et al. 2016b). In contrast, an excessive increase in symbiont density (driven by high enrichment with nutrients) may compromise the symbiosis, due to a strongly reduced translocation of organic carbon to the coral host (Cunning and Baker 2012; Ezzat et al. 2015). This effect is exacerbated under thermal stress. When symbiont respiration increases under increasing temperatures, the translocation of organic carbon to the host will be even further reduced to a critical level where the symbionts become commensalistic and maybe even parasitic instead of mutualistic (Wooldridge 2017; Baker et al. 2018). Such a shift towards a parasitic symbiosis has often been proposed to be a prelude to thermal bleaching (Cunning and Baker 2012; Wooldridge 2017; Baker et al. 2018; Morris et al. 2019).

It is not well known if corals can adapt to excessive enrichment with nutrients. Corals are well-suited to facilitate nutrient supply to their symbionts under oligotrophic conditions, through efficient ion carriers and storage systems (Davy et al. 2012) and even through endosymbiotic nitrogen-fixing bacteria (Rädecker et al. 2015). Pulses of nitrogen are effectively captured by the coral holobiont, at rates that exceed the requirement for maximal symbiont growth (D'Elia et al. 1983; Pernice et al. 2012). However, corals may be less effective in suppressing their nutrient acquisition systems under excess nutrient availability. Uncontrolled nutrient uptake can then destabilize the symbiosis and induce bleaching (Pogoreutz et al. 2017).

Over the last decade, it became apparent that sensitivity to thermal bleaching is largely determined by the stoichiometric ratio in which the key nutrient ions ammonium, nitrate (jointly referred to as DIN – Dissolved Inorganic Nitrogen), and phosphate (DIP – Dissolved Inorganic Phosphorous) occur, rather than by absolute levels of nutrient input. Under addition of DIN without a concurrent increase in DIP (which is a typical characteristic of manipulative nutrient enrichment – Fig. 1), phytoplankton cells multiply and consume DIP, resulting in reduced phosphate availability (Haese et al. 2007). Phosphate is required in corals in a specific ratio to nitrogen to maintain a stable photosymbiosis (Ezzat et al. 2016a). Phosphate shortage can affect the symbiosis in different ways. First, the algal symbionts may arrest their cell cycle, due to limited possibilities to synthesize DNA. Under these circumstances, the algae tend to become bigger and start storing their photosynthetic products instead of sharing them with the host (Wiedenmann et al. 2013). Second, under phosphate shortage, photosynthetic organisms will increasingly replace phospholipids with sulfoquinolone diacylglycerol (SQDG) to maintain the function of the

photosynthetic membranes and the embedded optical systems (Frentzen 2004). Wiedenmann et al. (2013) observed such an increase in SQDG in symbiodiniaceans *in hospite* under high nitrogen and low phosphorus conditions, which compromised the functioning of the light harvesting system. This malfunctioning reduces the photosynthetic efficiency and hence, the translocation of photosynthetic carbon to the coral host. The lack of translocation of organic carbon may fire back onto the symbiont, because the Carbon Concentrating Mechanism by which the coral supplies the symbiont with Dissolved Inorganic Carbon (DIC) largely depends on energy provided by symbiont photosynthesis (Wooldridge 2009). Carbon limitation of both the host and the symbiont, in combination with high illumination and increased temperature leads to impaired activity of PhotoSystem II and an associated excessive formation of ROS. ROS can damage both symbiont cells and host cells, and trigger the expelling or digestion of the symbionts by the coral host (Weis 2008; Wooldridge 2009; Fig. 4).

The vulnerability of corals to thermal bleaching particularly increases when unbalanced DIN enrichment occurs in the form of nitrate (Burkepile et al. 2019). Nitrate needs to be reduced to ammonium by the symbiotic algae through nitrate reductase (NR) and nitrite reductase (NiR) before it can be assimilated. The reduction process requires photosynthetic electrons and energy, and increases levels of Reactive Nitrogen Species (RNS) such as nitric oxide (NO) (Dagenais-Bellefeuille and Morse 2013; Ezzat et al. 2015; Fernandes de Barros Marangoni et al., 2020), thus further promoting oxidative stress in coral holobionts under thermal stress (Fernandes de Barros Marangoni et al., 2020; Fig. 4).

In contrast, the assimilation of ammonium does not incur a high metabolic cost to the holobiont. Both corals and symbionts use glutamine synthetase (GS) to incorporate ammonium directly in the amino acid glutamine, which can then be used for protein synthesis (Pernice et al. 2012; Dagenais-Bellefeuille and Morse 2013). Because of this direct benefit to protein turnover, moderate ammonium enrichment can even reduce physiological stress imposed by factors such as heat stress, elevated pCO_2 and pollution with dissolved organic carbon (Langdon and Atkinson 2005; Ezzat et al. 2019; Zhou et al. 2017; Bednarz et al. 2020; Fernandes de Barros Marangoni et al., 2020), provided that phosphate is not limiting (Ezzat et al. 2016a).

In summary, balanced enrichment with ammonium and phosphate can improve the resilience of corals to thermal bleaching, although this effect is not always visible on enriched natural reefs (e.g. Benkwitt et al. 2019). Likely, there is a threshold value above which symbiont densities become so high that the mutualistic character of the symbiosis

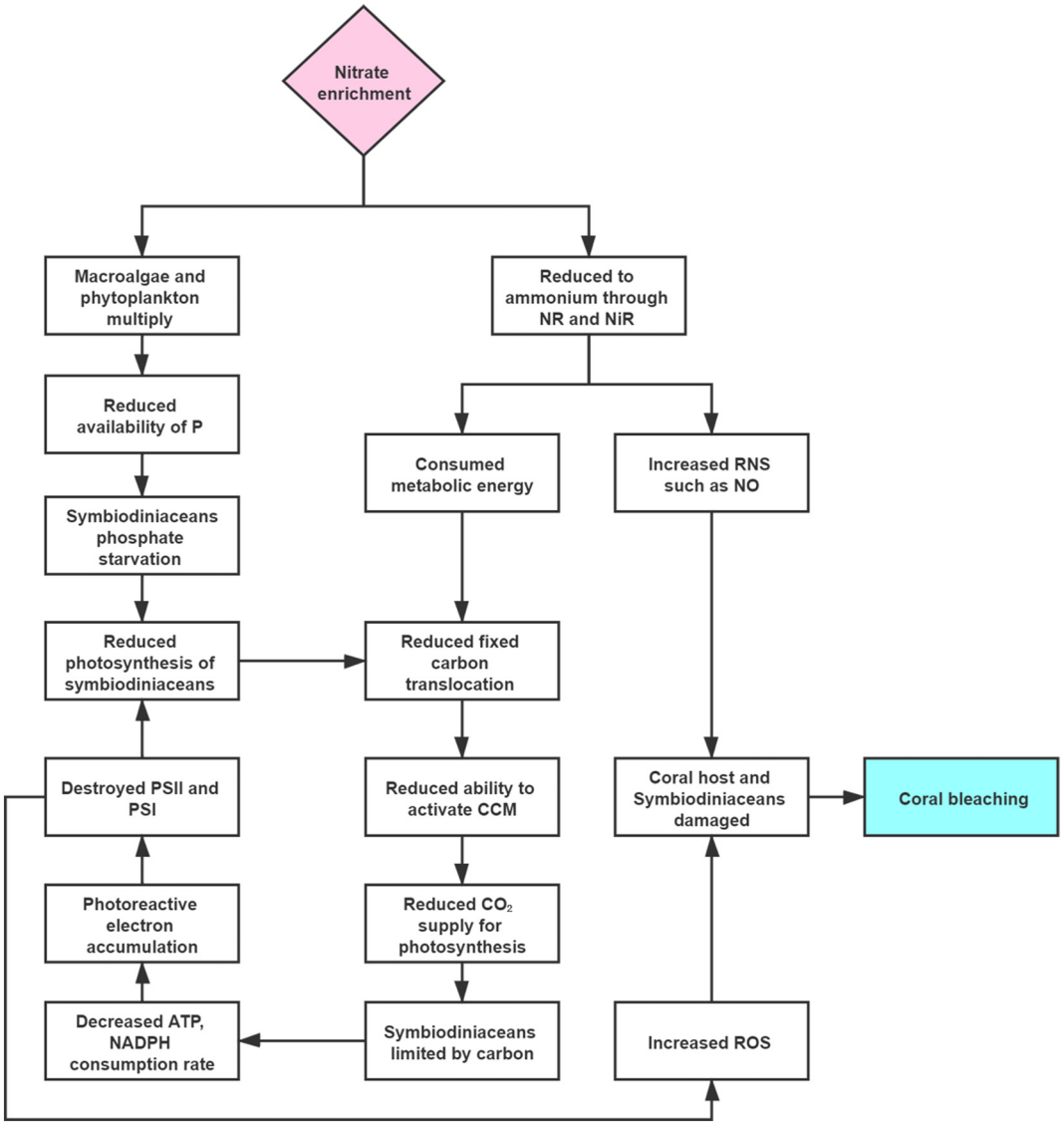

Fig. 4. Nitrate enrichment and coral bleaching.

will be compromised. Nitrogen enrichment without a concurrent enrichment in phosphate will increase the vulnerability of corals to thermal bleaching, because a shortage in phosphate causes unfavorable changes in the composition of the chloroplast membrane, thus compromising photosynthesis, reducing translocation of organic carbon and inducing oxidative stress. This effect is further exacerbated when nitrogen is supplied in the form of nitrate, because the reduction of nitrate to ammonium further increases the oxidative stress in the symbiotic algae through the formation of reactive nitrogen species (RNS), which act in synergy with the reactive oxygen species (ROS) produced through incomplete photosynthesis. Hence, high nitrate and low phosphate is the most detrimental combination of nutrient availability with respect to sensitivity to bleaching.

3.3. Nitrogen enrichment and coral calcification rate

Corals use calcium ions obtained from seawater and carbonate ions (CO_3^{2-}) accumulated through a CCM mechanism to build calcium carbonate skeletons (Ross et al. 2019). DIN and DIP are known to affect coral calcification. For many years, it was believed that enrichment with DIN and DIP increases the density of symbiotic algae in corals, which then leads to increased competition for DIC between symbionts and corals, resulting in reduced net coral calcification (Muscatine et al. 1989; Fabricius 2005). However, a meta-analysis of nutrient-enrichment studies on corals showed that calcification is differentially affected by nitrate and ammonium enrichment and that phosphate is stimulating calcification (Shantz and Burkepile 2014). Phosphate enrichment is assumed to relieve DIC limitation of calcification, because corals can incorporate $CaHPO_4$ in their skeleton instead of $CaCO_3$ (Dunn et al. 2012). Nitrate uptake by the symbionts reduces the translocation of photosynthetically acquired carbon to the coral host, because the reduction of nitrate to ammonium comes with a metabolic cost (Shantz and Burkepile 2014). This implies that nitrate affects calcification through a reduction in metabolic energy rather than competition for DIC, which is consistent with the observation that moderate enrichment with ammonium usually does not inhibit calcification (Fig. 5).

Experimental studies confirm the outcomes of the meta-analysis by Shantz and Burkepile (2014). Enrichment of aquarium corals with nitrate reduced calcification rates, whereas enrichment with ammonium did not (Fernandes de Barros Marangoni et al., 2020). Simultaneous enrichment with nitrate and phosphate also resulted in a decrease in the net calcification rate of individual and mixed coral communities (Silbiger et al. 2018). Mixed coral communities changed from net calcification to net dissolution under moderate and high nutrient conditions. This change was linked to a concurrent, nutrient-induced decrease in the community P:R ratio, which caused the microenvironment around the corals to become more acidic, thus reducing calcification (Silbiger et al. 2018). Their study implies that negative effects of nitrate on calcification are not alleviated by a concurrent addition of phosphate when the nitrate concentration is too high. In contrast to nitrate enrichment, ammonium enrichment at normal phosphorus levels promotes tissue growth of both corals and symbionts, and thus positively affects coral calcification (Table 1) (Shantz and Burkepile 2014; Guan et al. 2015; Johnson and Carpenter 2018; Fernandes de Barros Marangoni et al., 2020).

Nitrate-induced decline in calcification was found to interact synergistically with ocean acidification (DeCarlo et al. 2015; Prouty et al. 2017). Although it remains to be investigated to what extent this synergistic effect can be attributed to nitrate in particular, these authors proposed that nitrate enrichment might amplify the dissolution of coral skeleton caused by ocean acidification through a nutrient-driven increase in bioerosion (Fig. 5). Nutrient enrichment and high CO_2 promote the growth of macroalgae, which can stimulate the growth of bioeroders such as boring sponges through excretion of DOC (Müeller et al., 2014).

The bioerosion pathway is nowadays considered as a major route to reef decline (Schönberg et al. 2017). Skeletal damage caused by bioerosion can make corals more vulnerable to storms (Anthony 2016). It has been predicted that under future global warming, the intensity of reef erosion may increase by 45% as the frequency of storms increases by 10–40% (Emanuel 2013). Endolithic microorganisms including algae, fungi and cyanobacteria play a key role in nitrogen cycling in corals (Pernice et al. 2020). It can be proposed that the nitrogen balance between corals and endolithic microorganisms would be interrupted in the context of nitrogen enrichment. Some of the endolithic microorganisms can dissolve and weaken the coral $CaCO_3$ skeleton and therefore may play a significant role in reef's bioerosion (Pernice et al. 2020). However, experimental data are still required to investigate the bioerosion caused by endolithic microorganisms under the nitrogen enrichment conditions. To fully comprehend the causes of increased bioerosion, future studies on bioerosion and inorganic nutrients should differentiate between effects of the different nitrogen components. In addition, the role of phosphate in calcification and bioerosion should be further investigated. Phosphate has also been related to increased rates of bioerosion, as incorporation of $CaHPO_4$ may increase the porosity of the coral skeleton, making it more vulnerable to bioerosion (Dunn et al. 2012).

4. Strategies to conserve coral reefs

Management strategies and existing studies on coral reef conservation tend to focus on reversing the recession and restoring or preserving the ecosystem to its original states, functions, and service values (Rogers et al. 2015). However, increasing numbers of researchers have realized that it is impractical and impossible to restore coral reefs to the original state in the context of global climate change and the associated extinction of species (Abelson et al. 2016; van Oppen et al. 2017). Thus, the current goals for conservationists and managers are to help coral reefs adapt to the new environment and maintain their ecological service capabilities. To this end, we can reduce the current pollution status of those coral reef areas that have not been severely damaged yet and reduce the interferences from human activities (for example, overfishing and over-development of tourism) to let coral reef areas recover naturally. For degraded coral reef areas that cannot fully recover anymore through natural forces, it is of utmost importance to consider the functional biocomponents needed to maintain the primary ecosystem functions and implement appropriate active remediation measures. Hence, two main directions can be considered for future strategies: (1) passive restoration strategies - reducing and managing the nitrogen export from land to oceans and seas, protect the habitat and allow for natural recovery and improved ecological functioning including reef fish, and (2) active restoration strategies - direct interventions such as coral gardening and transplantation or manually removing over-proliferative macroalgae that aim to speed up recovery.

4.1. Reducing and managing the nitrogen export to oceans and seas

Land-based sources are important causes of nitrogen pollution in the sea and they differ among regions. For example, agriculture is considered a regional driver of nutrient enrichment in the Great Barrier Reef (De'ath and Fabricius 2010; Kroon et al. 2016; Packett 2017) and the South Pacific (Erler et al. 2018). In the Mexican Caribbean, high levels of nutrition in coral reef ecosystems are related to rapid coastal development and urbanization over the past four decades (Rioja-Nieto and Alvarez-Filip 2019). For the South China Sea, in addition to coastal development and agriculture, another driver that cannot be ignored is aquaculture (Herbeck et al. 2014 and 2020; Wang et al. 2020a; Wang et al. 2020b). For example, aquaculture around Hainan Island has caused nutrient pollution on adjacent reefs (Thomsen et al. 2020), contributing to a decline in coral cover from 32% in 2004 to 7% in 2018 according to the Status Report (Huang et al. 2020). Moreover, population growth

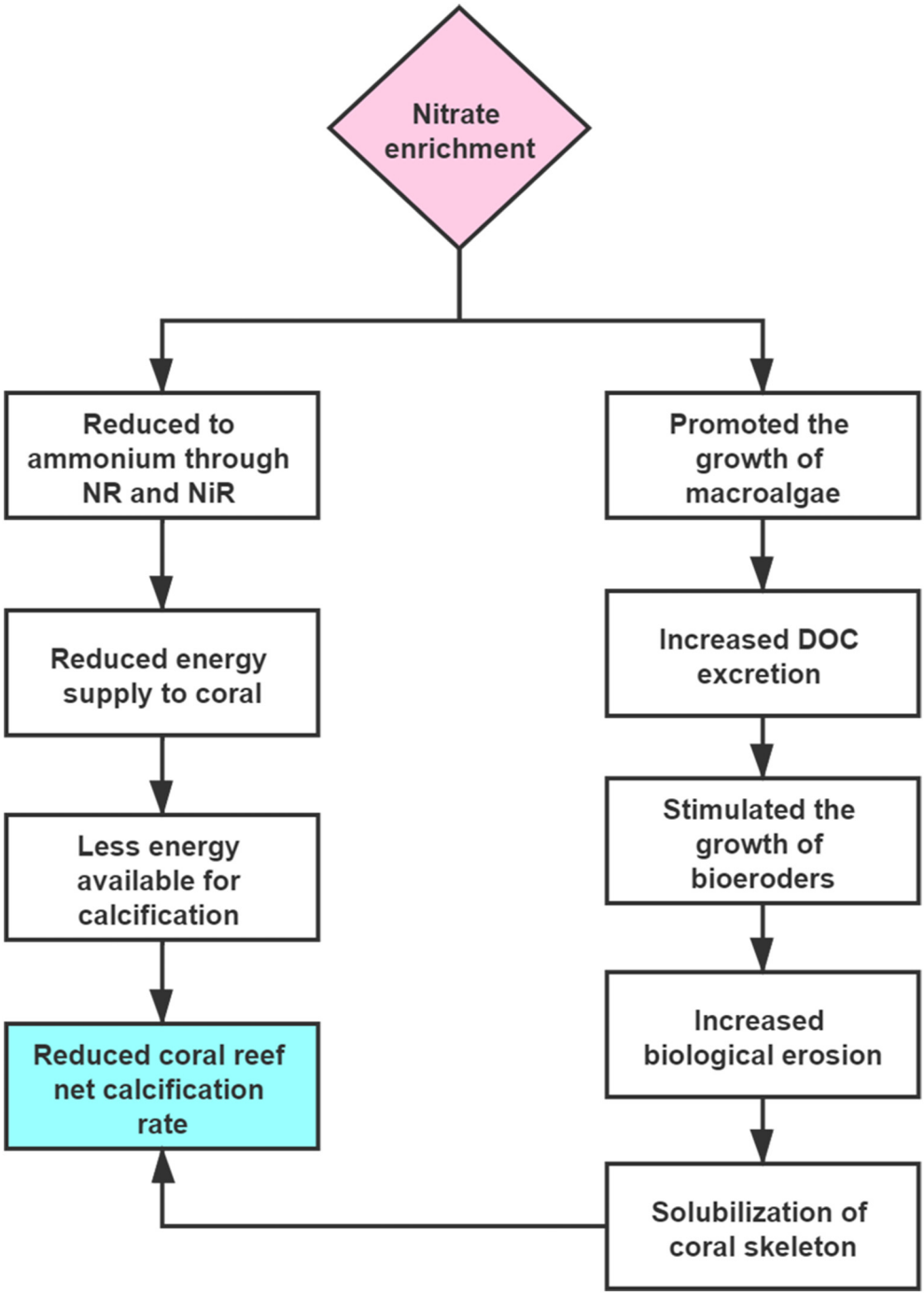

Fig. 5. Nitrate enrichment and coral calcification rate.

will exacerbate the development of coastal areas and the use of land and fertilizers. Effective management would facilitate the recovery of coral reefs. To this end, the Chinese government has prohibited fishing in the South China Sea from May to September in 2019 and 2020. Since 2019, Hainan Province has reduced coastal aquaculture and started to restore the mangrove forests that previously lined the coast. We believe that reducing nitrogen export to the sea from land-based sources, in particular nitrates, can positively influence the health of corals and coral reefs. Potential strategies to reduce land-based nitrogen export to the ocean include:

(1) Increasing nutrient use efficiencies in food production including establishing soil-crop system integrated management, testing soil for formulated fertilization, crop intercropping, reducing meat consumption and food waste, and recycling animal manure to farmland through integrating crop and livestock production (Chen et al. 2014; Zhang et al. 2015; Zhang et al. 2020b; Houlton et al. 2019; Strokal et al., 2020b). Recycling manure on land (instead of discharging to waters) is an important strategy for areas, such as China, where agriculture is a dominant source of nitrogen in rivers. Strokal et al. (2017), Misselbrook et al. (2020), Wang et al. (2020c), and Zhang et al. (2020a) analyzed the current situation in China and showed that this can decrease losses of agricultural nutrients to rivers and thus to oceans and seas and improve agricultural efficiency. The global population is projected to increase in the future, this will increase the

demand for food and thus make agricultural activities more intensive. Sustainable agricultural development is an effective way to guarantee the increase of food production while minimizing pollution to the environment. Currently, there are 2.5 billion smallholder farmers in the world. Most of these are localized in developing countries and manage 60% of the global arable land. The performance of these smallholder farmers not only determines their own livelihood, but also affects global food security and ecosystem health. The Science and Technology Backyard (STB) model is an illustrative example of sustainable agricultural practices with low environmental impacts. The STB model has been implemented in China where scientists and farmers work together towards sustainable production of agricultural products (Cui et al. 2018). Smallholder farmers that adopted the enhanced management practices from the STB model achieved higher food production yields while minimizing environmental impacts (Cui et al. 2018).

(2) Improving sanitation (more people being connected to sewage systems) together with improving wastewater treatment by implementing advanced technologies (Strokal et al. 2017 and 2020a; Tong et al. 2020). This strategy is especially important for urbanized deltas with poor sanitation and will reduce nitrogen discharge via wastewater treatment plants into rivers and seas. Examples of wastewater technologies are reverse osmosis, anammox and phosphate precipitation (Xie et al. 2016; Pérez-González et al. 2012; Ma et al. 2020). Changes in nitrogen and phosphorus levels in wastewater could result in a continuous increasing of N/P ratios (Tong et al. 2020). This imbalance may contribute to the development of harmful algae blooms in coastal waters ($N/P \gg 16$) (Beusen et al. 2013). Therefore, suitable N/P ratio ranges for the discharge of treated wastewater should be defined for future management.

(3) Restoring riparian vegetation and wetlands. Constructing a complex vegetation configuration pattern can improve the ecological functions of the riparian buffer zone, thereby increasing the purification capacity of surface runoff for total N, total P, available N, and available P, etc. (Hou et al. 2019). In addition, the restoration of wetlands can reduce nitrate concentrations and downstream suspended sediment loads (Czuba et al. 2018; McKergow et al. 2004).

(4) Making aquaculture more sustainable with precision feeding and by implementing multitrophic aquaculture approaches or advanced technologies to avoid losses of nutrients (Herbeck et al. 2020; Wang et al. 2020a). Aquaculture development by mangrove forest destruction should be banned. This strategy is important in areas where aquaculture is a dominant source of water pollution (e.g., Hainan, China).

(5) Land-sea integrated nutrient flow management is required for some areas with multiple intensive human activities. A model assessing river inputs of nutrients to oceans and seas will facilitate the control of coastal eutrophication. Strokal et al. (2016a, 2017, 2019), Li et al. (2019) and Wang et al. (2020c) established and developed the MARINA model that can quantitatively analyze the flux of land-based nutrients into river mouths. According to the nutrient tolerance (DIN < 9 μM and DIP < 0.33 μM, Fig. 2) of coral reef ecosystem (Duprey et al. 2016), the MARINA model could be used to calculate a coral reef safe threshold for river inputs of nutrients to coast. In addition, spatially explicit linked land-sea models can also potentially identify specific locations in catchments that are the largest potential drivers of coral reef shifts under different future scenarios (Delevaux et al. 2019).

In short, reducing and managing nitrogen export to seas requires the participation of multiple stakeholders (such as governments, scientists, managers, and the public) through sharing knowledge and power, formulating relevant laws and regulations and actively implementing them (Hughes et al. 2013; Hughes et al. 2017; Malone and Newton 2020). All stakeholders should cooperate to comprehensively detect, monitor and regulate nutrient input and its effects, and to establish models to rapidly evaluate the monitoring data. Furthermore, nutrient tolerance thresholds of different coral species and reef communities should be determined locally to enable tailored local management of terrestrial and atmospheric nitrogen input. This combination of strategies will enable management of marine nitrogen pollution on a regional as well as a global scale. Nitrogen enrichment is only one of the drivers of coral reef declining. Strategic conservation and management is required thorough investigation of the environmental and socioeconomic factors driving the persistence of reef-building coral assemblages by identifying key threats across multiple scales and strategic policy priorities (Darling et al. 2019).

4.2. Direct interventions and climate adaptation

Coral reef resilience varies from region to region (Roff and Mumby 2012). For areas with severely degraded coral reefs, simply eliminating local stress factors might not effectively restore coral reef ecosystems or may represent a strategy that requires a long recovery time (Done et al. 2010). Active restoration measures such as coral gardening (asexual coral propagation) and larval propagation can be considered after a site has been identified to support reef communities and when the causes of degradation (including control of nitrogen pollution as outlined in Section 4.1) have been stopped or effectively managed (Goergen and Gilliam 2018; Bostrom-Einarsson et al. 2020). Coral gardening has been successfully applied in the South China Sea, Caribbean, Western Atlantic and many other places, and is expected to contribute to the protection and restoration of coral reefs in areas where natural recovery is hindered by impaired coral recruitment (Lirman and Schopmeyer 2016; Zhang et al. 2016). However, it is unlikely to fully restore all coral reef ecosystem services (e.g., coastal protection) on a timeframe for local people to benefit.

In addition to mitigating local pressures such as nitrogen pollution and destructive fishing practices, the impact of climate change needs to be taken into account before active restoration is considered. Active restoration would not be effective in areas that are frequently hit by bleaching events, such as the Australasia and the Middle East (Hughes et al. 2018). Given the current prospects for future climate developments, new methods to enhance the resilience of newly formed reefs will be needed (van Oppen et al., 2017). Promising results in this respect are the successful application of assisted evolution to create corals that are better at withstanding heat waves (Buerger et al. 2020) and the translocation of corals growing in extreme habitats such as intertidal zones to more temperate environments (Schoepf et al. 2019). An interesting debate has started around the question whether the development towards “engineered coral reefs” is desirable (van Oppen et al., 2015). On the one hand, engineered reefs may be the only alternative if the Earth continues to warm up. On the other hand, the possibility to create engineered coral reefs may take away an important incentive to mitigate climate change.

5. Conclusion and outlook

In this paper, we first reviewed how marine nitrogen pollution is impacting coral reefs. Important sources of this pollution are food production (Herbeck et al. 2014; Hu et al. 2017; Wang et al. 2020a) and urbanization (Redding et al. 2013; Wong et al. 2017). The relative shares of the different sources in the total pollution vary from region to region. Second, we explored how different forms of nitrogen (e.g. nitrate and ammonium) and phosphorus affect corals. The possible mechanisms by which nitrate in particular cases increased coral disease incidence, coral bleaching, and decreased coral growth were summarized. We found that there is limited research on the genetic evolution of corals under long-term stress. Adaptations related to improved control of excessive nutrient influx by corals may be expected. Of particular interest

in this respect is the symbiosome (the host membrane that separates the host cell from the symbiont cell) where the carriers involved in nutrient transport to the symbiotic algae are located (Davy et al. 2012).

The differential effects of nitrate and ammonium require further study, in particular in relation to the occurrence and prevalence of coral diseases. Furthermore, it remains to be determined what happens under concurrent enrichment with nitrate and ammonium: will the benefits of ammonium outweigh the negative effects of nitrate or vice versa? The fact that nitrate uptake rates by freshly isolated symbiodiniaceans were not inhibited by the addition of ammonium (D'Elia et al. 1983) suggests that ammonium enrichment cannot alleviate nitrate-induced stress.

We identified a number of strategies to reduce the impact of nitrogen on corals under global change, such as improved efficiency of nutrient use in agriculture, improved sanitation and aquaculture systems, and halting global change itself by reducing carbon dioxide emissions. These strategies help to prevent further deterioration of corals, and to improve the status of the coral reefs. Currently, local management strategies should primarily focus on the reduction of nitrate input into coral reef environments and should strive for a balanced supply of moderate levels of phosphate and ammonium. The latter can be achieved by maintaining healthy coral reef fish populations, which can alleviate the competition from algae and warrant recycling of ammonium and phosphate in their environment through excretion of metabolic wastes (Shantz and Burkepile 2014).

We argue that research priority should be given to:

(1) Increasing the number of studies of the physiological and genetic adaptation of the coral host (epigenetic/genetic), the symbionts and the microbiome to adversity (e.g. temperature, eutrophication) to better understand the adaptation mechanisms and tolerance thresholds.
(2) Further understanding of the differential effects of enrichment with nitrate and ammonium, in particular in relation to coral disease and bioerosion. Moreover, the role of phosphate in calcification and bioerosion should be further investigated
(3) Development and improvement of models that determine the environmental carrying capacity for nutrient enrichment based on ecological tolerance thresholds of coastal coral reefs, while quantitatively analyzing the flux of land-based pollutants into coastal seas.
(4) Policy analyses to assess the effectiveness of measures to reduce future pollution of coral environments; a balance is needed between crop farming, aquaculture, animal husbandry, and household consumption as well as awareness raising and education about the sustainable development and protection of coral reef ecosystems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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